

Chapter 1

Introduction

*Why
study
statistics?*

Why study statistics?

Make a glance at a news paper. You may observe news that deal with:

- crime rates
- birth rates
- death rates
- average income, average rainfall
etc.

Why study statistics? (Contd. ...)

- The death rate in Ethiopia increased by 0.2 deaths per 1,000 inhabitants since 2020. In total, the death rate amounted to 6.78 deaths per 1,000 inhabitants in 2021.
- 45% of first marriages in Ethiopia end in divorce
[Jan 25, 2023]

Why study statistics? (Contd. ...)

- By a common definition, statistics consist of facts and figures.

ከሁለቱ መንግሥታዊ ባንኮች መካከል አንዱ የሆነው የኢትዮጵያ ልማት ባንክ በ2016 የሒሳብ ዓመት በዘጠኙ ወራት ከታክስ በፊት አራት ቢሊዮን ብር በማትረፍ የትርፍ ምጣኔውን ከ70 በመቶ በላይ ማሳደግ መቻሉ ተመለከተ።

Why study statistics? (Contd. ...)

ባንኩ የዘጠኙ ወራት የሥራ አፈጻጸሙን
አስመልክቶ ይፋ ባደረገው መረጃ በሦስተኛው ሩብ
ዓመት መጨረሻ ላይ ያስመዘገበው ትርፍ ከቀዳሚው
ዓመት ተመሳሳይ ወቅት ጋር ሲነፃፀር ከ1.65
ቢሊዮን ብር በላይ ብልጫ ያለው መሆኑንም ለማወቅ
ተችሏል።

Why study statistics? (Contd. ...)

በ2015 የሒሳብ ዓመት ባንኩ ከታክስ በፊት አስመዝግቦት የነበረው የትርፍ መጠን 2.35 ቢሊዮን ብር እንደነበር አይዘነጋም። ባንኩ በዘጠኙ ወራት ያስመገበው የትርፍ መጠን በባንኩ ታሪክ ከፍተኛ የዕድገት ምጣኔ ያሳየበት ነውም ተብሏል።

Why study statistics? (Contd. ...)

ኢትዮጵያ መንግሥቶች ክስተዳደር
ለ2016 በጀት ዓመት ለመንግሥት
ፕሮጀክቶች ግንባታ ከጠየቀው ሲሚንቶች
ማግኘት የቻለው 14 በመቶ ብቻ መሆኑን
ክስታወቀ::

Why study statistics? (Contd. ...)

በበጀት ዓመቱ ለመንገድ ፕሮጀክቶች ግንባታ ሥራ ከተጠየቀው 682,085 ቶን ሲሚንቶ ማግኘት የተቻለው 95,610 ቶን ወይም 14 በመቶ ብቻ መሆኑን፤ ለሕዝብ ተወካዮች ምክር ቤት የከተማ መሠረተ ልማትና ትራንስፖርት ጉዳዮች ቋሚ ኮሚቴ ጥር 13 ቀን 2016 ዓ.ም. ሪፖርት ያቀረቡት የኢትዮጵያ መንገዶች አስተዳደር ዋና ዳይሬክተር መሐመድ አብዱራህማን (ኢንጂነር)

Why study statistics? (Contd. ...)

በተመሳሳይ ክስተቶች ለግንባታ
ግብዓትነት ከጠየቀው 202,315,139
ሊትር ነዳጅ ውስጥ ማግኘት የቻለው
79,446,433 ሊትር ወይም 39 በመቶ ብቻ
እንደሆነ ገልጾዋል። Feb. 04, 2024

Why study statistics? (Contd. ...)

The increasing availability of computers and statistical software packages has enlarged the role of statistics as a tool for empirical research.

Why study statistics? (Contd. ...)

As a result statistics is used for research in almost all professions, from ***medicine to sport.***

Why study statistics? (Contd. ...)

It is in one form, or the other has become a common language by which agronomists, chemists, educators, *Programmers*, geneticists, political scientists, military strategists, medical researchers, sport managers, etc. read some portion of the world in which we live.

Why study statistics? (Contd. ...)

So why do we study Statistics?

Statistical knowledge helps you to use the *proper techniques to collect the data*, require precise calculations, and effectively present the conclusions.

Why study statistics? (Contd. ...)

Statistics is a vital process behind how we make discoveries in science, make decisions established on data, and make forecasts and predictions.

Why study statistics? (Contd. ...)

That is why we say:

“ mathematics or the body of mathematical applications called statistics is the language in which we read the universe.”

WHAT
IS
STATISTICS??

Definition:- Statistics is the branch of mathematics that deals with the scientific methods of collecting, organizing, and analyzing data so as to make conclusions, reasonable decisions and predictions on the basis of such analysis.

WHAT IS STATISTICS? (Contd. . . .)

The raw materials for any statistical analysis are **data** which are the numerical information from which we make analysis and interpretation.

STEPS OF STATISTICAL INVESTIGATION

STEPS OF STATISTICAL INVESTIGATION

The field of statistics is concerned with the following basic steps:

- The situation being investigated should be carefully and fully defined
- Proper collection of data

- Organization and classification of data
- Presentation of data
- Interpretation of data

IMPORTANCE OF STATISTICS

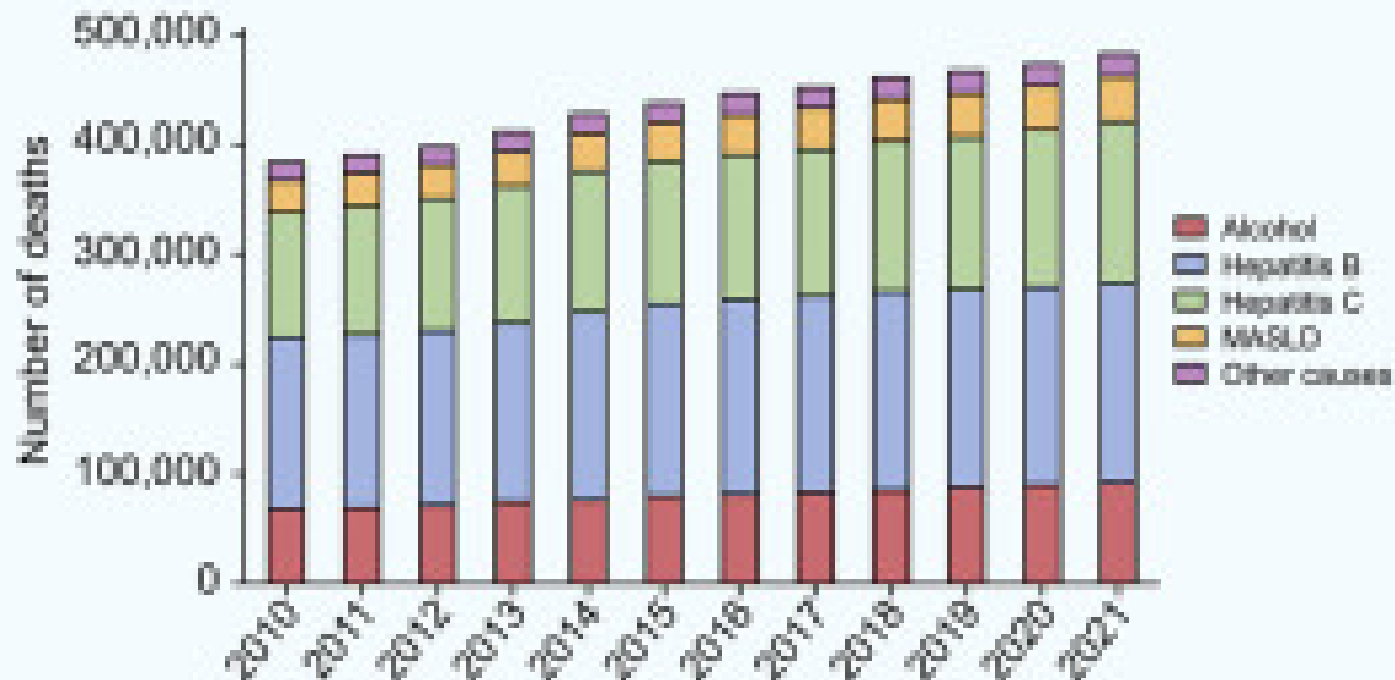
IMPORTANCE OF STATISTICS

- Statistics represents facts in the form of numerical data.
- Statistics condenses and summarizes a mass of data into few, presentable, understandable, and precise figures.

IMPORTANCE OF STATISTICS

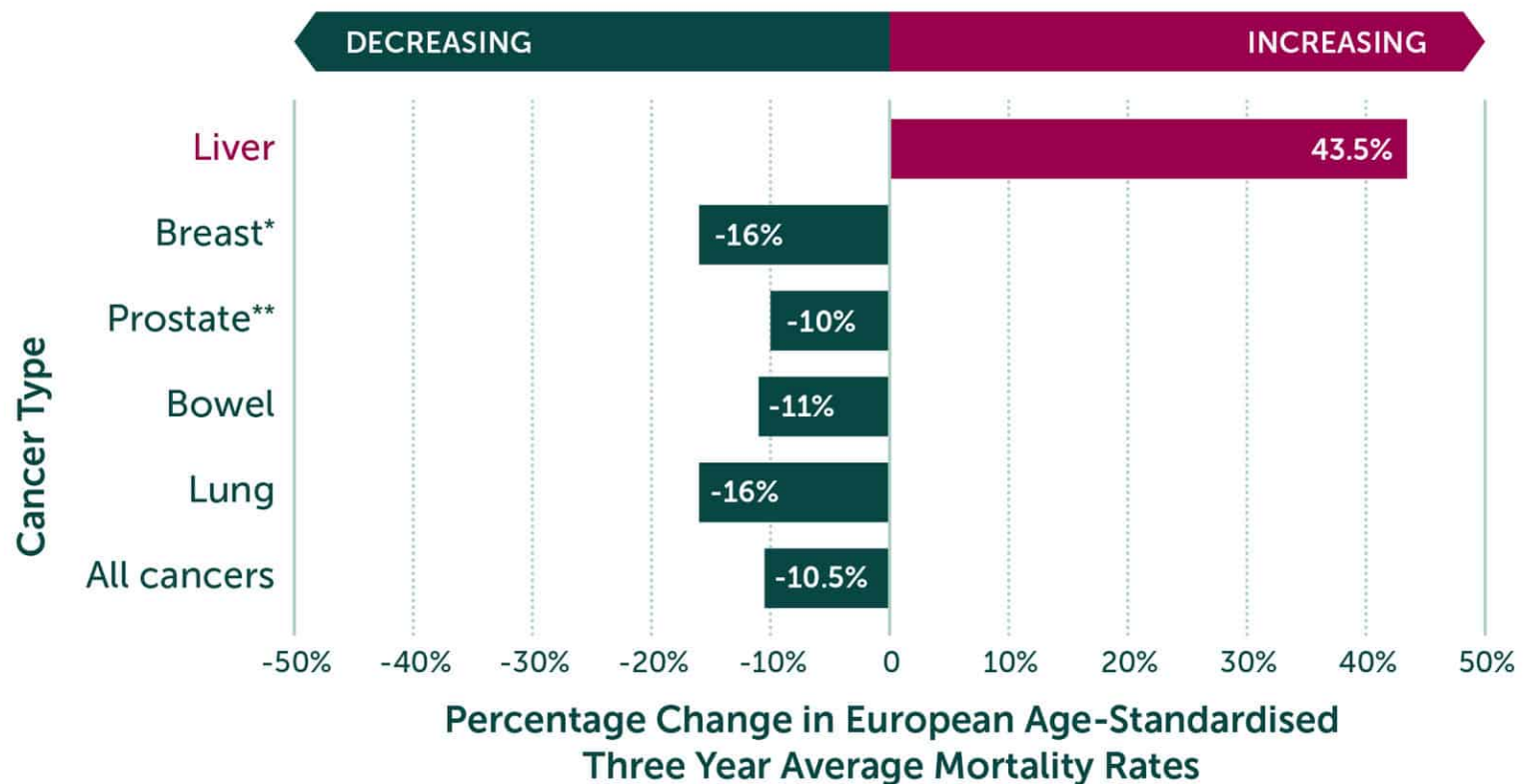
Liver cancer in 2021

- Absolute deaths increased by 25% from 2010 to 2021
- Global age-adjusted death rates from liver cancer declined, but increased in the Americas and Southeast Asia
- Age-adjusted death rates from MASLD increased sharply



IMPORTANCE OF STATISTICS (Contd. . .)

Cancer Mortality

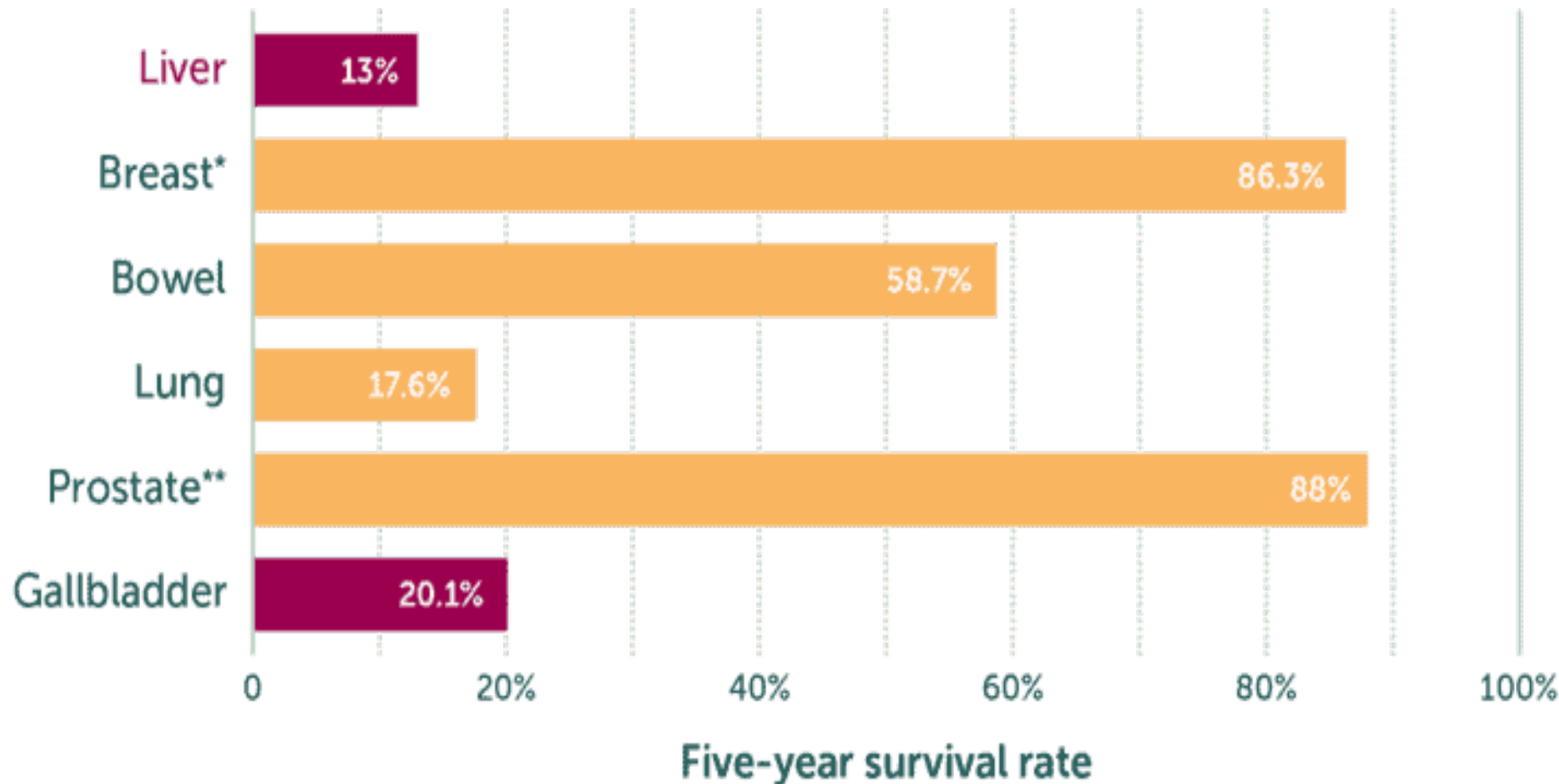


NB: Data has been combined for males and females.

*Women only **Men only

IMPORTANCE OF STATISTICS (Contd. . .)

Cancer Survival



*Women only **Men only

IMPORTANCE OF STATISTICS (Contd. . . .)

- Statistics facilitates classification and comparison of data.

በሞባይል የባንክ አገልግሎት በ2014 የሒሳብ ዓመት የነበረው የትራንዘክሽን መጠን 88 ሚሊዮን ሲሆን፣ በዚህ ትራንዘክሽን የተንቀሳቀሰው የገንዘብ መጠን 1.16 ትሪሊዮን ብር ነበር። በ**2011** የሒሳብ ዓመት በሞባይል ባንክ በ4.5 ሚሊዮን ትራንዘክሽን መጠን 16.8 ቢሊዮን ብር የገንዘብ ዝውውር ተደርጓል።

IMPORTANCE OF STATISTICS (Contd. . . .)

ይህ የገንዘብ መጠን በ2012 ወደ 68.3 ቢሊዮን ከዚያም

በ2013 ደግሞ 326.6 ቢሊዮን ማንቀሳቀስ መቻሉን ዕድገቱን

በተጨማሪም ያሳያል። ***February 4, 2024***

IMPORTANCE OF STATISTICS (Contd. . . .)

	Ethioia	United States
Believes crime was increasing for the past 3 years	72.22Ranked 6 th (11% more than US)	65.22 Ranked 44 th
Drugs (Annual cannabis use)	2.6%	13.7% (5 times more than Ethiopia)
Fear of crime (feel safe waking alone)	33.33	41.84 (26% more than Ethiopia)
Violent hate crime	13.89	32.55 (2 times more than Ethiopia)
Gun crime (per 100 residents)	0.4	88.8 (222 times more than Ethiopia)

IMPORTANCE OF STATISTICS (Contd. . . .)

- Statistics helps the central management and government in formulating policies.
- Statistics helps in predicting future trends.

IMPORTANCE OF STATISTICS (Contd. . . .)

Planned Condominium Housing Unit Construction, 2006 – 2010

	Studio	1 bed room	2 bed rooms	3 bed rooms	Total units
Addis Ababa	36,750	71,750	52,500	14,000	175,000
Other Regional States	38,850	75,850	55,500	14,800	185,000
Total	75,600	147,600	180,000	28,800	306,000
% distribution	21%	41%	30%	8%	100%



IMPORTANCE OF STATISTICS (Contd. . . .)

Source: adapted from Dolicho, E. (2006) Condominium, A New Housing Development Approach for Addressing the Low Income Housing Problem of Addis Ababa, Ethiopia (Final Report). ICHUD, IHS, June 2006, Rotterdam, The Netherlands.

IMPORTANCE OF STATISTICS (Contd. . . .)

According to the United States Department of Agriculture, for the year 2022-23(Oct-Sep), Ethiopia is expected to produce 8.25 million 60-kilogram bags of coffee (495,000 MT).

IMPORTANCE OF STATISTICS (Contd. . . .)

The USDA forecasts Ethiopia's 2023–24 coffee year exports to be up by 600 MT. The Coffee market is expected to show a volume growth, at home of 0.7% in 2025.

LIMITATIONS OF STATISTICS

• Statistics does not deal with individual values.

i.e. By statistics we mean aggregates of facts and placed in relation to each other.

It means statistics deals with only aggregates of facts or items and it does not recognize any individual item.

LIMITATIONS OF STATISTICS

❑ *Statistics does not deal with qualitative characteristics.*

The statistical methods don't study the nature of phenomenon which cannot be expressed in quantitative terms. Such phenomena cannot be a part of the study of statistics.

These include honesty, kindness, beauty etc.

Sometimes it needs conversion of qualitative data into quantitative data. We mean: Gender, Marital status, Qualification , etc.

- *Statistical conclusions are not universally true.*

So, we can say that we can check the data and procedures of its approaching to conclusions. But these data may have been collected by inexperienced persons or they may have been dishonest or biased.

As it is a delicate science and can be easily misused by an untrustworthy person. So, data must be used with a caution. Otherwise, results may prove to be disastrous.

LIMITATIONS OF STATISTICS

- *Statistical interpretation requires a high degree of skill and understanding of the subject.*

LIMITATIONS OF STATISTICS

Statistics deal with figures which are innocent in themselves and can be easily manipulated or distorted by people for their selfish motives. Therefore, it is a dangerous tool in the hands of a non-expert.

LIMITATIONS OF STATISTICS

It is hence important that the user of the statistical methods has sound knowledge of the subject .

LIMITATIONS OF STATISTICS

- *Statistics can be misused.*

Misleading statistics are data points or sets that lead readers to believe something that isn't true.

LIMITATIONS OF STATISTICS

Statistics are occasionally misused to persuade, influence and sell. Misuse can also result from mistake of analysis that result in poor decision and failed strategies.

Consider the following example:

LIMITATIONS OF STATISTICS

In 2007, toothpaste company *Colgate* ran an advertising stating that 80% of dentists recommend their product. Based on the promotion, many shoppers assumed Colgate was the best choice for their dental health.

LIMITATIONS OF STATISTICS

But this wasn't necessarily true.

In reality, this is a famous example of misleading statistics.

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TERMINOLOGIES

TERMINOLOGIES

Definition 1:- A population in statistical terms is the set of elements that belong to the entire group.

Definition 2:- A summary measure that describes any given characteristic of the population is known as a parameter.

TERMINOLOGIES (Contd. . . .)

- **Definition 3:-** A sample is a portion of the total population that is considered for study and analysis.
- **Definition 4:-** A summary measure that describes a characteristic of the given sample is known as a statistic.

TYPES OF STATISTICS

Definition 1:- The phase of statistics, which seeks only to describe and analyze a sample without drawing any conclusion about a population is called ***descriptive statistics.***

Definition2:- The phase of statistics, which deals with drawing important conclusions or predictions about a population from analysis of a sample is called **inferential statistics**.

45% of first marriages in Ethiopia end in divorce within 30 years, and two-thirds of women who divorce do so within the first 5 years of marriage.

The two main factors that may have an impact on the risk of divorce in Ethiopia are: early age of first marriage, and childlessness within the first marriage. A total of 8757 women of reproductive age (15-49) were analysed.

National Family and Fertility Survey conducted by the Government of Ethiopia (1990)

End of Chapter 1



Any Questions?

